

$$q(\mathbf{x}_{1:T}|\mathbf{x}_0) = \prod_{t=1}^T q(\mathbf{x}_t|\mathbf{x}_{t-1})$$

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I})$$

$$\alpha_t = 1 - \beta_t$$

$$\mathbf{x}_t = \sqrt{\alpha_t}\mathbf{x}_{t-1} + \sqrt{1 - \alpha_t}\mathbf{z}_{t-1}, \quad \mathbf{z}_{t-1} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

$$\mathbf{x}_{t-1} = \sqrt{\alpha_{t-1}}\mathbf{x}_{t-2} + \sqrt{1 - \alpha_{t-1}}\mathbf{z}_{t-2}, \quad \mathbf{z}_{t-2} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

$$\begin{aligned} \mathbf{x}_t &= \sqrt{\alpha_t} \left(\sqrt{\alpha_{t-1}}\mathbf{x}_{t-2} + \sqrt{1 - \alpha_{t-1}}\mathbf{z}_{t-2} \right) + \sqrt{1 - \alpha_t}\mathbf{z}_{t-1} \\ &= \sqrt{\alpha_t\alpha_{t-1}}\mathbf{x}_{t-2} + \sqrt{\alpha_t(1 - \alpha_{t-1})}\mathbf{z}_{t-2} + \sqrt{1 - \alpha_t}\mathbf{z}_{t-1} \end{aligned}$$

$$\sigma_{\text{new}} = \sqrt{\sigma_1^2 + \sigma_2^2}$$

$$\begin{aligned} \text{Variance} &= \alpha_t(1 - \alpha_{t-1}) + (1 - \alpha_t) \\ &= \alpha_t - \alpha_t\alpha_{t-1} + 1 - \alpha_t \\ &= 1 - \alpha_t\alpha_{t-1} \end{aligned}$$

$$\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$$

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1 - \bar{\alpha}_t)\mathbf{I})$$

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t}\boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \frac{q(\mathbf{x}_t|\mathbf{x}_{t-1}, \mathbf{x}_0)q(\mathbf{x}_{t-1}|\mathbf{x}_0)}{q(\mathbf{x}_t|\mathbf{x}_0)}$$

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \propto \exp\left(-\frac{1}{2}\left[\frac{(\mathbf{x}_t - \sqrt{\alpha_t}\mathbf{x}_{t-1})^2}{\beta_t} + \frac{(\mathbf{x}_{t-1} - \sqrt{\bar{\alpha}_{t-1}}\mathbf{x}_0)^2}{1 - \bar{\alpha}_{t-1}} - \frac{(\mathbf{x}_t - \sqrt{\bar{\alpha}_t}\mathbf{x}_0)^2}{1 - \bar{\alpha}_t}\right]\right)$$

$$\text{Exponent} = -\frac{1}{2}\left[\frac{\mathbf{x}_t^2 - 2\sqrt{\alpha_t}\mathbf{x}_t\mathbf{x}_{t-1} + \alpha_t\mathbf{x}_{t-1}^2}{\beta_t} + \frac{\mathbf{x}_{t-1}^2 - 2\sqrt{\bar{\alpha}_{t-1}}\mathbf{x}_0\mathbf{x}_{t-1} + \bar{\alpha}_{t-1}\mathbf{x}_0^2}{1 - \bar{\alpha}_{t-1}} - C_1\right]$$

$$\text{Exponent} = -\frac{1}{2}\left[\left(\frac{\alpha_t}{\beta_t} + \frac{1}{1 - \bar{\alpha}_{t-1}}\right)\mathbf{x}_{t-1}^2 - 2\left(\frac{\sqrt{\alpha_t}\mathbf{x}_t}{\beta_t} + \frac{\sqrt{\bar{\alpha}_{t-1}}\mathbf{x}_0}{1 - \bar{\alpha}_{t-1}}\right)\mathbf{x}_{t-1} + C_2\right]$$

$$a = \frac{\alpha_t}{\beta_t} + \frac{1}{1 - \bar{\alpha}_{t-1}} = \frac{\alpha_t(1 - \bar{\alpha}_{t-1}) + \beta_t}{\beta_t(1 - \bar{\alpha}_{t-1})} = \frac{1 - \bar{\alpha}_t}{\beta_t(1 - \bar{\alpha}_{t-1})}$$

$$b = -2\left(\frac{\sqrt{\alpha_t}\mathbf{x}_t}{\beta_t} + \frac{\sqrt{\bar{\alpha}_{t-1}}\mathbf{x}_0}{1 - \bar{\alpha}_{t-1}}\right)$$

$$-\frac{(\mathbf{x}_{t-1} - \tilde{\boldsymbol{\mu}}_t)^2}{2\tilde{\beta}_t} = -\frac{1}{2}\left(\frac{1}{\tilde{\beta}_t}\mathbf{x}_{t-1}^2 - \frac{2\tilde{\boldsymbol{\mu}}_t}{\tilde{\beta}_t}\mathbf{x}_{t-1}\right) + C_3$$

$$\frac{1}{\tilde{\beta}_t} = a \implies \tilde{\beta}_t = \frac{\beta_t(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \quad -\frac{2\tilde{\boldsymbol{\mu}}_t}{\tilde{\beta}_t} = b \implies \tilde{\boldsymbol{\mu}}_t = -\frac{\tilde{\beta}_t b}{2}$$

$$\tilde{\boldsymbol{\mu}}_t = \frac{\beta_t(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \left(\frac{\sqrt{\alpha_t}\mathbf{x}_t}{\beta_t} + \frac{\sqrt{\bar{\alpha}_{t-1}}\mathbf{x}_0}{1 - \bar{\alpha}_{t-1}}\right)$$

$$= \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t}\mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t}\mathbf{x}_0$$

$$\mathbf{x}_0 = \frac{1}{\sqrt{\bar{\alpha}_t}}(\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t}\mathbf{z}_t)$$

$$\tilde{\boldsymbol{\mu}}_t = \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t}\mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t} \cdot \left[\frac{1}{\sqrt{\bar{\alpha}_t}}(\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t}\mathbf{z}_t)\right]$$

$$= \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t}\mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{(1 - \bar{\alpha}_t)\sqrt{\alpha_t}\sqrt{\bar{\alpha}_{t-1}}}\mathbf{x}_t - \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t\sqrt{1 - \bar{\alpha}_t}}{(1 - \bar{\alpha}_t)\sqrt{\alpha_t}\sqrt{\bar{\alpha}_{t-1}}}\mathbf{z}_t$$

$$= \left[\frac{\alpha_t(1 - \bar{\alpha}_{t-1}) + \beta_t}{(1 - \bar{\alpha}_t)\sqrt{\alpha_t}}\right]\mathbf{x}_t - \frac{\beta_t}{\sqrt{\alpha_t}\sqrt{1 - \bar{\alpha}_{t-1}}}\mathbf{z}_t$$

$$= \frac{1}{\sqrt{\alpha_t}}\mathbf{x}_t - \frac{\beta_t}{\sqrt{\alpha_t}\sqrt{1 - \bar{\alpha}_t}}\mathbf{z}_t$$

$$= \frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}}\mathbf{z}_t\right)$$

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}\left(\mathbf{x}_{t-1}; \tilde{\boldsymbol{\mu}}_t, \tilde{\beta}_t\mathbf{I}\right) \quad \mathbf{z}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

$$\begin{aligned}
L_{\text{VLB}} &= \mathbb{E}_{q(\mathbf{x}_{0:T})} \left[\log \frac{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}{p_\theta(\mathbf{x}_{0:T})} \right] \\
&= \mathbb{E}_q \left[\log \frac{\prod_{t=1}^T q(\mathbf{x}_t|\mathbf{x}_{t-1})}{p_\theta(\mathbf{x}_T) \prod_{t=1}^T p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)} \right] \\
&= \mathbb{E}_q \left[-\log p_\theta(\mathbf{x}_T) + \sum_{t=1}^T \log \frac{q(\mathbf{x}_t|\mathbf{x}_{t-1})}{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)} \right] \\
&= \mathbb{E}_q \left[-\log p_\theta(\mathbf{x}_T) + \sum_{t=2}^T \log \frac{q(\mathbf{x}_t|\mathbf{x}_{t-1})}{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)} + \log \frac{q(\mathbf{x}_1|\mathbf{x}_0)}{p_\theta(\mathbf{x}_0|\mathbf{x}_1)} \right] \\
&= \mathbb{E}_q \left[-\log p_\theta(\mathbf{x}_T) + \sum_{t=2}^T \log \left(\frac{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)}{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)} \cdot \frac{q(\mathbf{x}_t|\mathbf{x}_0)}{q(\mathbf{x}_{t-1}|\mathbf{x}_0)} \right) + \log \frac{q(\mathbf{x}_1|\mathbf{x}_0)}{p_\theta(\mathbf{x}_0|\mathbf{x}_1)} \right] \\
&= \mathbb{E}_q \left[-\log p_\theta(\mathbf{x}_T) + \sum_{t=2}^T \log \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)}{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)} + \sum_{t=2}^T \log \frac{q(\mathbf{x}_t|\mathbf{x}_0)}{q(\mathbf{x}_{t-1}|\mathbf{x}_0)} + \log \frac{q(\mathbf{x}_1|\mathbf{x}_0)}{p_\theta(\mathbf{x}_0|\mathbf{x}_1)} \right] \\
&= \mathbb{E}_q \left[-\log p_\theta(\mathbf{x}_T) + \sum_{t=2}^T \log \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)}{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)} + \log \frac{q(\mathbf{x}_T|\mathbf{x}_0)}{q(\mathbf{x}_1|\mathbf{x}_0)} + \log \frac{q(\mathbf{x}_1|\mathbf{x}_0)}{p_\theta(\mathbf{x}_0|\mathbf{x}_1)} \right] \\
&= \mathbb{E}_q \left[\log \frac{q(\mathbf{x}_T|\mathbf{x}_0)}{p_\theta(\mathbf{x}_T)} + \sum_{t=2}^T \log \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)}{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)} - \log p_\theta(\mathbf{x}_0|\mathbf{x}_1) \right] \\
&= \mathbb{E}_q \left[\underbrace{D_{\text{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \parallel p_\theta(\mathbf{x}_T))}_{L_T} + \underbrace{\sum_{t=2}^T D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))}_{L_{t-1}} - \underbrace{\log p_\theta(\mathbf{x}_0|\mathbf{x}_1)}_{L_0} \right]
\end{aligned}$$